Static Scattering Theory Notes

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To gain information about the structure of matter, ranging from elementary particles to solids, one studies the scattering of particles (electrons, neutrons, He atoms, photons) (Fig 1)

The particle incident on the target will be represented by a wave packet. This must be large compared to the dimensions of the scatterer, but small compared to the remaining spatial dimensions, in order not to overlap simultaneously with the scatterer and the detector. Its width in momentum space must be sufficiently narrow, so that the spreading during the experiment is negligible. After the interaction, the wave function consists of an unscattered part propagating in the forward direction as well as a scattered part. In what follows, we consider a single scattering center described by a potential $V(\mathbf{r})$, which is taken to be situated at the origin of coordinates.

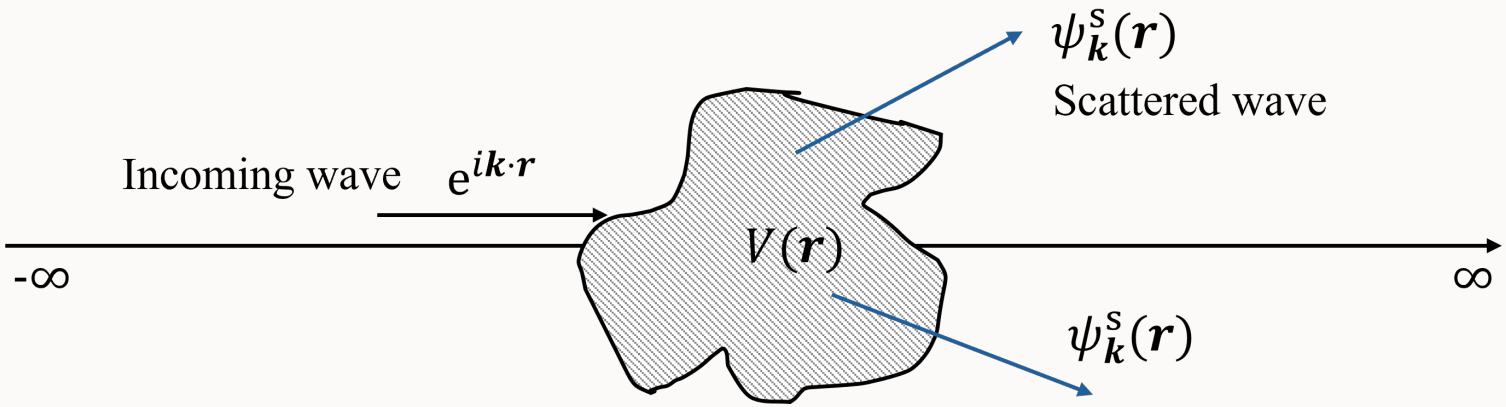

Figure 1: Scattering Process Sketch

Within the potential, the corresponding time-independent Schr$\ddot{o}$inger equation is,

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})\right]\psi_{\mathbf{k}}(\mathbf{r}) = E_{\mathbf{k}}\psi_{\mathbf{k}}(\mathbf{r}) \tag{1}$$

where $\psi_{\mathbf{k}}$ is the exact eigenstate, $E_{\mathbf{k}}$ is set to be $\frac{\hbar^2 k^2}{2m}$ (Elastic scattering).

1 The Lippmann-Schwinger Equation: Position Representation

The equation can be solved via Green's function method. (1) can be written as,

$$\left(\nabla^2 + k^2\right)\psi_{\mathbf{k}}(\mathbf{r}) = \frac{2m}{\hbar^2}V(\mathbf{r})\psi_{\mathbf{k}}(\mathbf{r}) \tag{2}$$

Remark (Green's function method) Given a linear differential operator $L(x)$ acting on $u(x)$, gives $f(x)$,

$$Lu(x) = f(x)$$

The Green's function of $L(x)$ is,

$$LG(x-s) = \delta(x-s)$$

One has

$$\begin{aligned}\int LG(x-s)f(s)ds &= \int \delta(x-s)f(s)d(s) = f(x) \\ &= L\int G(x-s)f(s)ds\end{aligned}$$

which implies

$$u(x) = \int G(x-s)f(s)ds$$

Apply this to (2). The Green's function of $\left(\nabla^2 + k^2\right)$ is,

$$\left(\nabla^2 + k^2\right) G_0(\mathbf{r}-\mathbf{r}') = \delta^{(3)}(\mathbf{r}-\mathbf{r}') \tag{3}$$

Thus, the solution is

$$\psi^s_{\mathbf{k}}(\mathbf{r}) = \frac{2m}{\hbar^2}\int d^3\mathbf{r}' G_0(\mathbf{r}-\mathbf{r}')V(\mathbf{r}')\psi_{\mathbf{k}}(\mathbf{r}')$$

It should be noted that the solution one got here is the particular solution, which originates from the source term. It is thus called the scattered wave. The complete solution is composed of the particular solution and the complementary solution, referring to the solution of the corresponding homogeneous equation. In our case, the complementary solution is $Ae^{i\mathbf{k}\cdot\mathbf{r}} + Be^{-i\mathbf{k}\cdot\mathbf{r}}$. $e^{-i\mathbf{k}\cdot\mathbf{r}}$ propagates from infinity towards minus infinity. Only $e^{i\mathbf{k}\cdot\mathbf{r}}$ is kept. The complete solution is

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} + \frac{2m}{\hbar^2}\int d^3\mathbf{r}' G_0(\mathbf{r}-\mathbf{r}')V(\mathbf{r}')\psi_{\mathbf{k}}(\mathbf{r}') \tag{4}$$

which is the **Lippmann-Schwinger equation** in coordinate space.

1.1 Solve for Green's Function

A common technique used in solving such problems is the Fourier transform. Consider the Fourier transform of $G_0(\mathbf{r})$,

$$\tilde{G}_0(\mathbf{q}) = \int d^3\mathbf{r}\, G_0(\mathbf{r})e^{-i\mathbf{q}\cdot\mathbf{r}} \quad G_0(\mathbf{r}) = \int \frac{d^3\mathbf{q}}{(2\pi)^3}\tilde{G}_0(\mathbf{q})e^{i\mathbf{q}\cdot\mathbf{r}}$$

Do a Fourier transform on both sides of (3), we have

$$\begin{aligned}
\int d^3\mathbf{r}\left(\nabla^2+k^2\right)G_0(\mathbf{r})e^{-i\mathbf{q}\cdot\mathbf{r}} &= \int d^3\mathbf{r}\,\delta^{(3)}(\mathbf{r})e^{-i\mathbf{q}\cdot\mathbf{r}} \\
k^2\tilde{G}_0(\mathbf{q})+\int d^3\mathbf{r}\left(\nabla^2 G_0(\mathbf{r})\right)e^{-i\mathbf{q}\cdot\mathbf{r}} &= 1 \\
k^2\tilde{G}_0(\mathbf{q})+F\left[\nabla^2 G_0(\mathbf{r})\right] &= 1 \\
k^2\tilde{G}_0(\mathbf{q})+(i\mathbf{q})^2F\left[G_0(\mathbf{r})\right] &= 1 \\
\tilde{G}_0(\mathbf{q}) &= \frac{1}{-q^2+k^2}
\end{aligned}$$

Thus,

$$\begin{aligned}
G_0(\mathbf{r}) &= \int \frac{d^3\mathbf{q}}{(2\pi)^3}\frac{e^{i\mathbf{q}\cdot\mathbf{r}}}{(-q^2+k^2)} \\
&= \frac{1}{4\pi^2}\int_0^\infty dq\int_{-1}^1 d\cos\theta\frac{q^2e^{iqr\cos\theta}}{-q^2+k^2} \\
&= \frac{1}{4\pi^2 ir}\int_0^\infty dq\frac{q(e^{iqr}-e^{-iqr})}{-q^2+k^2} \\
&= \frac{1}{4\pi^2 ir}\left[\int_0^\infty dq\frac{qe^{iqr}}{-q^2+k^2}+\int_\infty^0 dq\frac{qe^{-iqr}}{-q^2+k^2}\right] \\
&= \frac{1}{4\pi^2 ir}\int_{-\infty}^\infty dq\frac{qe^{iqr}}{-q^2+k^2} \\
&= \frac{1}{4\pi^2 ir}\lim_{\epsilon\to 0^+}\int_{-\infty}^\infty dq\frac{-qe^{iqr}}{(q-k-i\epsilon)(q+k+i\epsilon)}
\end{aligned}$$

The integrand decays to 0 in the upper half-plane. Choose the upper half-plane as the contour (Fig 2).

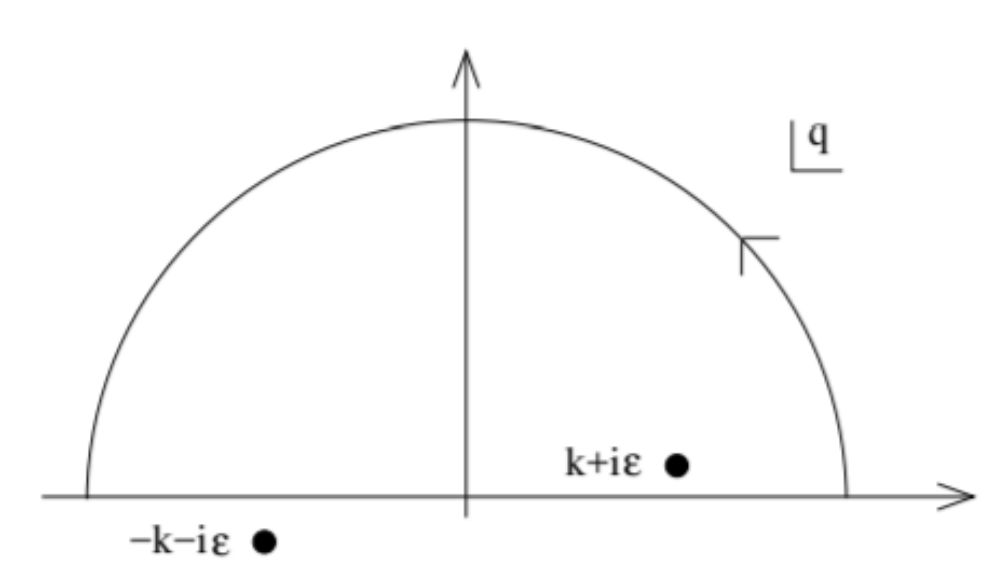

Figure 2: Contour integral

$$
\begin{aligned}
G_0^+(\mathbf{r}) &= \frac{2\pi i}{4\pi^2 i r} \lim_{\epsilon \to 0^+} \frac{-(k+i\epsilon)e^{i(k+i\epsilon)r}}{2(k+i\epsilon)} \\
&= -\frac{1}{4\pi}\frac{e^{ikr}}{r}
\end{aligned}
$$

(4) becomes

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} - \frac{m}{2\pi\hbar^2}\int d^3\mathbf{r}'\frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}V(\mathbf{r}')\psi_{\mathbf{k}}(\mathbf{r}') \tag{5}$$

If we take $k = k - i\epsilon$, we will have $G_0^-(\mathbf{r}-\mathbf{r}') = \frac{1}{4\pi}\frac{e^{-ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}$. The propagation of this wave is toward the potential, which is not the desired solution.

$$G_0^{\pm}(\mathbf{r}-\mathbf{r}') = \mp\frac{1}{4\pi}\frac{e^{\pm ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}$$

Then, the wave function becomes

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} - \frac{m}{2\pi\hbar^2}\int d^3\mathbf{r}'\frac{e^{\pm ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}V(\mathbf{r}')\psi_{\mathbf{k}}(\mathbf{r}') \tag{6}$$

2 Scattering Amplitude

Under far-field approximation, i.e. the observer is far from the scatterer, or we say the asymptotic behavior at $r \to \infty$. The variable in the wave function, $\mathbf{r}$, is where the wave function is measured by a detector. On the other hand, $\mathbf{r}'$ is the integral variable, also the variable of $V(\mathbf{r}')$. Since the potential is localized, the integral is within the potential. This implies $r \gg r'$. Thus,

$$|\mathbf{r}-\mathbf{r}'| = \sqrt{r^2 - 2\mathbf{r}\cdot\mathbf{r}' + r'^2} \approx r - \frac{\mathbf{r}\cdot\mathbf{r}'}{r}$$

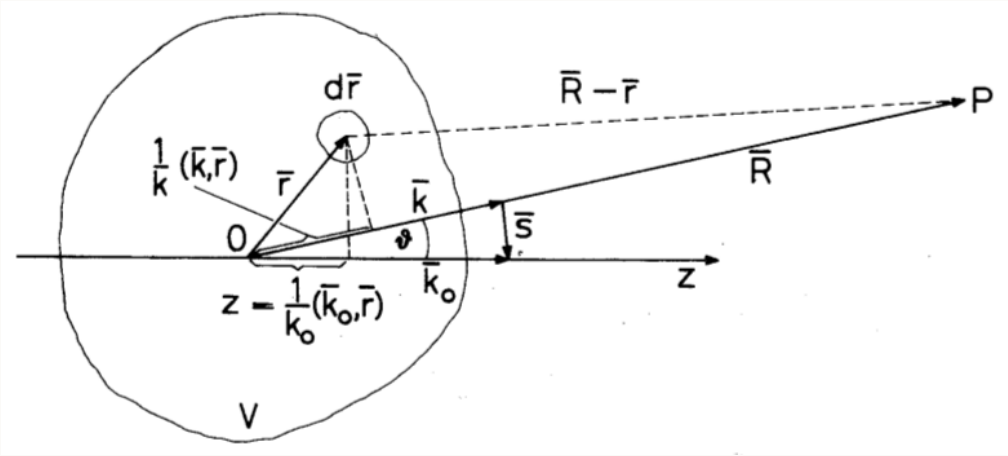

Figure 3: Illustration of the description of electron scattering; bars indicate vectors.

Then,

$$\psi_{\mathbf{k}}(\mathbf{r}) \approx e^{i\mathbf{k}\cdot\mathbf{r}} - \frac{m}{2\pi\hbar^2}\int d^3\mathbf{r}' \frac{e^{ik(r-\frac{\mathbf{r}\cdot\mathbf{r}'}{r})}}{r-\frac{\mathbf{r}\cdot\mathbf{r}'}{r}} V(\mathbf{r}')\psi_{\mathbf{k}}(\mathbf{r}')$$

$$\approx e^{i\mathbf{k}\cdot\mathbf{r}} - \left[\frac{m}{2\pi\hbar^2}\int d^3\mathbf{r}' e^{-i\mathbf{k}'\cdot\mathbf{r}'} V(\mathbf{r}')\psi_{\mathbf{k}}(\mathbf{r}')\right]\frac{e^{ikr}}{r} \tag{7}$$

$$\equiv e^{i\mathbf{k}\cdot\mathbf{r}} + f(\theta,\phi)\frac{e^{ikr}}{r} \tag{8}$$

where $\mathbf{k}' = \frac{k\mathbf{r}}{r}$. Since elastic scattering was assumed, one has $k' = k$. $\mathbf{r}/r$ is the unit vector along the scattering direction. $\mathbf{k}'$ is, therefore, the momentum of scattered wave being detected.

$$f(\theta,\phi) = -\frac{m}{2\pi\hbar^2}\int d^3\mathbf{r}' e^{-i\mathbf{k}'\cdot\mathbf{r}'} V(\mathbf{r}')\psi_{\mathbf{k}}(\mathbf{r}')$$

is the scattering amplitude.

3 The Lippmann-Schwinger Equation: A More General Discussion

Instead of working under position representation, one can choose to work with vectors.

$$(E_{\mathbf{k}} - H_0)\,|\psi_{\mathbf{k}}\rangle = V\,|\psi_{\mathbf{k}}\rangle$$

Since the eigenstates are the superposition of the incident plane wave $|\mathbf{k}\rangle$ and the scattered state, one can introduce the plane wave part to simplify the problem.

$$\begin{aligned}(E_{\mathbf{k}} - H_0)\,|\mathbf{k}\rangle &= 0\\ (E_{\mathbf{k}} - H_0)(|\psi_{\mathbf{k}}\rangle - |\mathbf{k}\rangle) &= V\,|\psi_{\mathbf{k}}\rangle\end{aligned} \tag{9}$$

Introducing $i\varepsilon(\varepsilon > 0)$ and taking the inverse,

$$\left|\psi_{\mathbf{k}}^{\pm}\right\rangle = |\mathbf{k}\rangle + \frac{1}{E_{\mathbf{k}} - H_0 \pm i\varepsilon} V\,\left|\psi_{\mathbf{k}}^{\pm}\right\rangle \tag{10}$$

which is the **Lippmann-Schwinger equation**.

Define the free-particle Green's operator

$$G_0^{\pm}(E) = \frac{1}{E_{\mathbf{k}} - H_0 \pm i\varepsilon}$$

Obviously, $\left(G_0^+\right)^\dagger = G_0^-$. Claim that $G_0^\pm(\mathbf{r}-\mathbf{r}')$ is related to $G_0^\pm(E)$ evaluated at $\mathbf{r}, \mathbf{r}'$. To show this

$$\begin{aligned}
\langle\mathbf{r}| \frac{1}{E_\mathbf{k} - H_0 \pm i\varepsilon} |\mathbf{r}'\rangle &= \int d^3\mathbf{q} \, \langle\mathbf{r}| \frac{1}{E_\mathbf{k} - H_0 \pm i\varepsilon} |\mathbf{q}\rangle \langle\mathbf{q}|\mathbf{r}'\rangle \\
&= \frac{2m}{\hbar^2} \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{1}{k^2 - q^2 \pm i\varepsilon} e^{i\mathbf{q}\cdot(\mathbf{r}-\mathbf{r}')} \\
&= \mp \frac{2m}{\hbar^2} \frac{1}{4\pi} \frac{e^{\pm ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} \\
&= \frac{2m}{\hbar^2} G_0^\pm(\mathbf{r}-\mathbf{r}')
\end{aligned}$$

3.1 The Full Green's Operator

Now, we introduce the full Green's operator. Rearrange eqn (8):

$$\begin{aligned}
(E_\mathbf{k} - H_0) \left|\psi_\mathbf{k}^\pm\right\rangle - V \left|\psi_\mathbf{k}^\pm\right\rangle &= (E_\mathbf{k} - H_0) |\mathbf{k}\rangle \\
(E_\mathbf{k} - H) \left|\psi_\mathbf{k}^\pm\right\rangle &= (E_\mathbf{k} - H) |\mathbf{k}\rangle + V |\mathbf{k}\rangle
\end{aligned}$$

One gets the Lippmann-Schwinger equation in the form of full Green's operator.

$$\left|\psi_\mathbf{k}^\pm\right\rangle = |\mathbf{k}\rangle + \frac{1}{E_\mathbf{k} - H \pm i\varepsilon} V |\mathbf{k}\rangle \tag{11}$$

where

$$G^\pm(E) = \frac{1}{E - H \pm i\varepsilon}$$

is the full Green's operator. Notice that the right side contains only $|\mathbf{k}\rangle$, which no longer seems to be an iterative equation. Unfortunately, the full Green's operator is unknown. And $|\mathbf{k}\rangle$ is not an eigenstate of $G^{\pm}(\mathbf{k})$. It leads us to nowhere.

3.2 The Dyson Identity

By plugging eqn (10) into eqn (9),

$$\left|\psi_{\mathbf{k}}^{\pm}\right\rangle = |\mathbf{k}\rangle + G_0^{\pm}(E)V\,|\mathbf{k}\rangle + G_0^{\pm}(E)VG^{\pm}(E)V\,|\mathbf{k}\rangle$$

Compare it with eqn (10), one gets the **Dyson identity**

$$G^{\pm}(E) = G_0^{\pm}(E) + G_0^{\pm}(E)VG^{\pm}(E) \tag{12}$$

Similarly, from eqn (9),

$$|\mathbf{k}\rangle = \left|\psi_{\mathbf{k}}^{\pm}\right\rangle - G_0^{\pm}(E)V\left|\psi_{\mathbf{k}}^{\pm}\right\rangle$$

Plugging into eqn (10),

$$\left|\psi_{\mathbf{k}}^{\pm}\right\rangle = |\mathbf{k}\rangle + G^{\pm}(E)V\left|\psi_{\mathbf{k}}^{\pm}\right\rangle - G^{\pm}(E)VG_0^{\pm}(E)V\left|\psi_{\mathbf{k}}^{\pm}\right\rangle$$

Thus,

$$G^{\pm}(E) = G_0^{\pm}(E) + G^{\pm}(E)VG_0^{\pm}(E) \tag{13}$$

which is an equivalent form of (11).

3.3 From the Perspective of Coordinate Space

Back to eqn (4) and replace the plane wave with $\varphi_{\mathbf{k}}(\mathbf{r})$, which is the solution of the free-particle Hamiltonian.

$$\psi_{\mathbf{k}}^{\pm}(\mathbf{r}) = \varphi_{\mathbf{k}}(\mathbf{r}) + \frac{2m}{\hbar^2}\int d^3\mathbf{r}' G_0^{\pm}(\mathbf{r}-\mathbf{r}')V(\mathbf{r}')\psi_{\mathbf{k}}^{\pm}(\mathbf{r}') \tag{14}$$

Similar to what we have done above,

$$\begin{aligned}
(\nabla^2 + k^2)[\psi_{\mathbf{k}}^{\pm}(\mathbf{r}) - \varphi_{\mathbf{k}}(\mathbf{r})] &= \frac{2m}{\hbar^2}V(\mathbf{r})\psi_{\mathbf{k}}^{\pm}(\mathbf{r}) \\
\left[\nabla^2 + k^2 - \frac{2m}{\hbar^2}V(\mathbf{r})\right]\psi_{\mathbf{k}}^{\pm}(\mathbf{r}) &= \left[\nabla^2 + k^2 - \frac{2m}{\hbar^2}V(\mathbf{r})\right]\varphi_{\mathbf{k}}(\mathbf{r}) + \frac{2m}{\hbar^2}V(\mathbf{r})\varphi_{\mathbf{k}}(\mathbf{r})
\end{aligned}$$

Select the full Green's function to satisfy

$$\begin{aligned}
\left[\nabla^2 + k^2 - \frac{2m}{\hbar^2}V(\mathbf{r})\right]G^{\pm}(\mathbf{r}-\mathbf{r}') &= \delta^{(3)}(\mathbf{r}-\mathbf{r}') \\
\nabla^2 + k^2 - \frac{2m}{\hbar^2}V(\mathbf{r}) &\equiv \mathcal{L}(\mathbf{r})
\end{aligned}$$

Multiply both sides by $\frac{2m}{\hbar^2}V(\mathbf{r}')\varphi_{\mathbf{k}}(\mathbf{r}')$, and integrate

$$\begin{aligned}
\int d^3\mathbf{r}'\mathcal{L}(\mathbf{r})G^{\pm}(\mathbf{r}-\mathbf{r}')\frac{2m}{\hbar^2}V(\mathbf{r}')\varphi_{\mathbf{k}}(\mathbf{r}') &= \frac{2m}{\hbar^2}V(\mathbf{r})\varphi_{\mathbf{k}}(\mathbf{r}) \\
&= \mathcal{L}(\mathbf{r})\int d^3\mathbf{r}' G^{\pm}(\mathbf{r}-\mathbf{r}')\frac{2m}{\hbar^2}V(\mathbf{r}')\varphi_{\mathbf{k}}(\mathbf{r}')
\end{aligned}$$

which implies

$$\psi_{\mathbf{k}}^{\pm}(\mathbf{r}) = \varphi_{\mathbf{k}}(\mathbf{r}) + \int d^3\mathbf{r}' G^{\pm}(\mathbf{r}-\mathbf{r}') \frac{2m}{\hbar^2} V(\mathbf{r}') \varphi_{\mathbf{k}}(\mathbf{r}') \tag{15}$$

Substitute (14) into (13), one gets

$$G^{\pm}(\mathbf{r}-\mathbf{r}') = G_0^{\pm}(\mathbf{r}-\mathbf{r}') + \int d^3\mathbf{r}' G_0^{\pm}(\mathbf{r}-\mathbf{r}') V(\mathbf{r}') G^{\pm}(\mathbf{r}-\mathbf{r}') \tag{16}$$

4 Scattering Cross Section

Definition 4.1 *The differential cross section $d\sigma/d\Omega$ is defined as, the ratio of the average number of particles $dN/d\Omega$ entering per unit solid angle per unit time in the direction (θ, ϕ) at a distance to the average particle flux density $\bar{\mathbf{J}}_0$ at the incident angle. Therefore, $\sigma(\theta, \phi)$ has the dimension of area.*

$\hat{\mathbf{n}}_{\mathbf{r}}$ is the unit vector along (θ, ϕ), then under far-field approximation

$$d\sigma/d\Omega = |f(\theta, \phi)|^2$$

5 T Operator & S Operator

To solve the cross section, one needs to look at the scattering amplitude,

$$\begin{aligned} f(\theta,\phi) &= -\frac{m}{2\pi\hbar^2}\int d^3\mathbf{r}' e^{-i\mathbf{k}_f\cdot\mathbf{r}'}V(\mathbf{r}')\psi_{\mathbf{k}}^{+}(\mathbf{r}') \\ &= -\frac{(2\pi)^{3/2}m}{2\pi\hbar^2}\int d^3\mathbf{r}' \left\langle \mathbf{k}_f|\mathbf{r}'\right\rangle V(\mathbf{r}')\left\langle \mathbf{r}'\middle|\psi_{\mathbf{k}}^{+}\right\rangle \\ &= -\frac{\sqrt{2\pi}m}{\hbar^2}\int d^3\mathbf{r}' \left\langle \mathbf{k}_f|\mathbf{r}'\right\rangle \left\langle \mathbf{r}'\right| V \left|\psi_{\mathbf{k}}^{+}\right\rangle \\ &= -\frac{\sqrt{2\pi}m}{\hbar^2}\left\langle \mathbf{k}_f\right| V \left|\psi_{\mathbf{k}}^{+}\right\rangle \end{aligned}$$

To write the scattering amplitude as a matrix element of an operator, define T operator

$$T^{\pm}\left|\mathbf{k}_i\right\rangle = V\left|\psi_{\mathbf{k}_i}^{\pm}\right\rangle$$

Thus, the scattering amplitude can be written as

$$f(\theta,\phi) = -\frac{\sqrt{2\pi}m}{\hbar^2}\left\langle \mathbf{k}_f\right| T^{+}\left|\mathbf{k}_i\right\rangle$$

Times V on the left side of two forms of Lippmann-Schwinger equation,

$$\begin{aligned} V\left|\psi_{\mathbf{k}}^{\pm}\right\rangle &= V\left|\mathbf{k}\right\rangle + V\frac{1}{E_{\mathbf{k}}-H_0\pm i\varepsilon}V\left|\psi_{\mathbf{k}}^{\pm}\right\rangle \\ V\left|\psi_{\mathbf{k}}^{\pm}\right\rangle &= V\left|\mathbf{k}\right\rangle + V\frac{1}{E_{\mathbf{k}}-H\pm i\varepsilon}V\left|\mathbf{k}\right\rangle \end{aligned}$$

One has

$$T^{\pm} = V + VG_0^{\pm}(E)T^{\pm} \tag{17}$$

$$T^{\pm} = V + VG^{\pm}(E)V \tag{18}$$

Since $G_0^{\pm}(E)$ and $G^{\pm}(E)$ are functions of E, $T^{\pm}$ are also functions of E. The above two are the Lippmann-Schwinger equation of T operator.

Easily, one gets

$$G_0^{\pm}T^{\pm} = G^{\pm}V$$

$$T^{\pm}G_0^{\pm} = VG^{\pm}$$

$$G^{\pm} = G_0^{\pm} + G_0^{\pm}T^{\pm}G_0^{\pm}$$

Introduce two Møller operators, $\Omega^{\pm}$. They directly transform the free-particle state to the scattered state:

$$\Omega^{\pm} \left|\mathbf{k}_i\right\rangle = \left|\psi_{\mathbf{k}_i}^{\pm}\right\rangle$$

According to the Lippmann-Schwinger equation and the completeness of free-particle states, Møller operators are definite,

$$\left\langle\mathbf{k}'\right| \Omega^{\pm} \left|\mathbf{k}\right\rangle = \left\langle\mathbf{k}'\middle|\psi_{\mathbf{k}}^{\pm}\right\rangle$$

$$\Omega^{\pm} = \int \frac{d^3\mathbf{k}'}{(2\pi)^3}\frac{d^3\mathbf{k}}{(2\pi)^3} \left|\mathbf{k}'\right\rangle \left\langle\mathbf{k}'\middle|\psi_{\mathbf{k}}^{\pm}\right\rangle \left\langle\mathbf{k}\right|$$

$$\Omega^{\pm} = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \left|\psi_{\mathbf{k}}^{\pm}\right\rangle \left\langle\mathbf{k}\right|$$

Notice that

$$
\begin{aligned}
\left(\Omega^{\pm}\right)^{\dagger} &= \int \frac{d^3\mathbf{k}}{(2\pi)^3} \left|\mathbf{k}\right\rangle \left\langle \psi_{\mathbf{k}}^{\pm}\right| \\
\left(\Omega^{\pm}\right)^{\dagger} \Omega^{\pm} &= \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{d^3\mathbf{k}'}{(2\pi)^3} \left|\mathbf{k}\right\rangle \left\langle \psi_{\mathbf{k}}^{\pm} \middle| \psi_{\mathbf{k}'}^{\pm}\right\rangle \left\langle \mathbf{k}'\right| \\
&= \int \frac{d^3\mathbf{k}}{(2\pi)^3} \left|\mathbf{k}\right\rangle \left\langle \mathbf{k}\right| = \mathbb{1}
\end{aligned}
$$

Because the free-particle states are complete, if the potential satisfies some restrictions, the eigenstates of the full Hamiltonian are also complete. But the scattered states are not necessarily complete, because H may have bound states $|\phi_i\rangle$:

$$
\begin{aligned}
\int \frac{d^3\mathbf{k}}{(2\pi)^3} \left|\psi_{\mathbf{k}}^{\pm}\right\rangle \left\langle \psi_{\mathbf{k}}^{\pm}\right| + \int \frac{d^3n}{(2\pi)^3} \left|\varphi_n\right\rangle \left\langle \varphi_n\right| &= \mathbb{1} \\
\sum_{\mathbf{k}} \left|\psi_{\mathbf{k}}^{\pm}\right\rangle \left\langle \psi_{\mathbf{k}}^{\pm}\right| + \sum_i \left|\varphi_i\right\rangle \left\langle \varphi_i\right| &= \mathbb{1}
\end{aligned}
$$

Not sure if the integral form is correct. Define $B \equiv \sum_i \left|\varphi_i\right\rangle \left\langle \varphi_i\right| = \int \frac{d^3n}{(2\pi)^3} \left|\varphi_n\right\rangle \left\langle \varphi_n\right|$

$$
\Omega^{\pm} \left(\Omega^{\pm}\right)^{\dagger} = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \left|\psi_{\mathbf{k}}^{\pm}\right\rangle \left\langle \psi_{\mathbf{k}}^{\pm}\right| = \mathbb{1} - B
$$

Thus, the Møller operator is an equidistant operator when H has bound states, and is a unitary operator when H has no bound states.

Møller operators can be used to construct a unitary operator S:

$$S = \left(\Omega^-\right)^\dagger \Omega^+$$

Then

$$\begin{aligned}
S^\dagger S &= \left(\Omega^+\right)^\dagger \Omega^- \left(\Omega^-\right)^\dagger \Omega^+ \\
&= \left(\Omega^+\right)^\dagger (\mathbb{1} - B)\, \Omega^+ \\
&= \mathbb{1} - \left(\Omega^+\right)^\dagger B \Omega^+ \\
&= \mathbb{1} - \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{d^3\mathbf{k}'}{(2\pi)^3} \frac{d^3 n}{(2\pi)^3} \left|\mathbf{k}\right\rangle \left\langle \psi_{\mathbf{k}}^+ \middle| \varphi_n \right\rangle \left\langle \varphi_n \middle| \psi_{\mathbf{k}'}^+ \right\rangle \left\langle \mathbf{k}'\right| \\
&= \mathbb{1}
\end{aligned}$$

$\left|\varphi_n\right\rangle$ and $\left|\psi_{\mathbf{k}}^+\right\rangle$ should be orthogonal. Similarly,

$$SS^\dagger = \left(\Omega^-\right)^\dagger \Omega^+ \left(\Omega^+\right)^\dagger \Omega^- = \mathbb{1}$$

Consider the matrix element of S:

$$
\begin{aligned}
S_{fi} &= \langle \mathbf{k}_f | S | \mathbf{k}_i \rangle = \langle \mathbf{k}_f | \left(\Omega^-\right)^\dagger \Omega^+ | \mathbf{k}_i \rangle = \left\langle \psi^-_{\mathbf{k}_f} \middle| \psi^+_{\mathbf{k}_i} \right\rangle \\
&= \left[\langle \mathbf{k}_f | + \langle \mathbf{k}_f | V \frac{1}{E_f - H + i\varepsilon} \right] \left| \psi^+_{\mathbf{k}_i} \right\rangle \\
&= \left\langle \mathbf{k}_f \middle| \psi^+_{\mathbf{k}_i} \right\rangle + \frac{1}{E_f - E_i + i\varepsilon} \langle \mathbf{k}_f | V \left| \psi^+_{\mathbf{k}_i} \right\rangle \\
&= \langle \mathbf{k}_f | \left[| \mathbf{k}_i \rangle + \frac{1}{E_i - H_0 + i\varepsilon} V \left| \psi^+_{\mathbf{k}_i} \right\rangle \right] + \frac{1}{E_f - E_i + i\varepsilon} \langle \mathbf{k}_f | V \left| \psi^+_{\mathbf{k}_i} \right\rangle \\
&= \langle \mathbf{k}_f | \mathbf{k}_i \rangle + \left[\frac{1}{E_i - E_f + i\varepsilon} + \frac{1}{E_f - E_i + i\varepsilon} \right] \langle \mathbf{k}_f | V \left| \psi^+_{\mathbf{k}_i} \right\rangle \\
&= \delta_{fi} - \frac{2i\varepsilon}{(E_i - E_f)^2 + \varepsilon^2} \langle \mathbf{k}_f | T^+ | \mathbf{k}_i \rangle \\
&= \delta_{fi} - 2\pi i \delta(E_i - E_f) T^+_{fi}
\end{aligned}
$$

Thus,

$$
S = \mathbb{1} - 2\pi i \delta(E_i - E_f) T^+
$$

not sure